Empirical computation

1. Inputs

Site contrasts and estimated variances

2. Working normal model

Flat reference prior on mean and SD

Condition on positive site variances

3. Posterior integration

Integrate the common mean analytically

Adaptive quadrature over $\mathrm{SD} \geq 0$

4. Reliability profile

Evaluate I, $\bar{S}$, H and J_{eff}

V is fixed by the variance profile

5. Report

Posterior means and 95% credible intervals

Compare summaries for the intended goal

OCRS: course passing

32 school-by-subject sites

Mean [95% credible interval]

$I = 0.671\ [0.497, 0.812]$

$H = 0.025\ [0.019, 0.029]$

$\bar{S} = 0.663$

$J_{\text{eff}} = 31.2;\ V = 0.237$

$\Pr(I > 0.5) = 0.973$

Plug-in $I = 0.691$

Credits in year 4

$I = 0.318\ [0.022, 0.597]$

Reported values use adaptive quadrature.